

Missing Figure Completion

COMPLETE LEARNING GUIDE

What You'll Learn

Missing-figure completion questions show one large figure with a chunk cut out — usually a clean rectangular or wedge-shaped notch — and ask which of several candidate pieces correctly fills the gap. Unlike figure-matrix questions, there is no grid of separate cells; instead a single figure's internal pattern, symmetry, lines, or borders must be continued smoothly across the missing region. This sub-module builds a complete method: continuing an internal pattern, using symmetry to derive the mirror-image piece, tracing lines and curves across the gap at the correct position and angle, completing radial (wheel-like) and border patterns, and using outline-fit and detail elimination as final filters before selecting an answer.

1. The Core Skill: Continuing the Figure's Internal Pattern

A missing-piece completion question shows one large figure with a chunk cut out and asks which candidate piece correctly fills that gap — the guiding principle is that the missing piece must continue whatever pattern, lines, or design elements run through the rest of the figure, not merely fit the gap's outline shape.

- A figure is a grid of horizontal stripes, with one rectangular notch missing from the middle. Even though several candidate pieces are the exact right rectangular size and shape to physically fit the notch, only the one showing horizontal stripes at the correct spacing and alignment actually completes the pattern correctly.

2. Symmetry-Based Completion

When the overall figure has an obvious line of symmetry and the missing piece is on one side of that line, the correct piece is simply the mirror image of the corresponding intact region on the OTHER side of the symmetry line — locate that mirroring region first before considering any answer option.

- A butterfly-like figure is symmetric left-to-right, with a notch missing from the lower-right wing. The corresponding lower-LEFT wing (intact) shows a specific curved pattern with 2 small circles. The correct missing piece must be the mirror image of that exact lower-left pattern — 2 small circles in mirrored positions, following the same curve reflected left-to-right.

3. Line and Curve Continuation Across the Gap

For figures built from continuous lines or curves that pass through the missing region, trace each line from where it enters the gap on one side and determine exactly where and at what angle it must exit on the other side — the correct piece must connect these entry and exit points smoothly, matching both position and angle, not just position alone.

- A diagonal line enters the missing notch from the top-left corner of the gap, heading down-and-right at a consistent angle. It must exit the gap's boundary at the bottom-right corner, continuing at that SAME angle. A candidate piece where the line bends partway through, or exits at the wrong point, is incorrect even if the line generally 'looks like it connects.'

4. Radial or Repeating Pattern Completion

In circular or wheel-like figures made of a repeating wedge pattern around a center point, the missing wedge must repeat the exact same internal design as every other wedge in the figure, rotated to the correct angular position — count the total number of wedges and the angle each spans to determine precisely what the missing wedge's design and orientation must be.

- A circular figure is divided into 8 equal wedges (45° each), 7 of which show an identical small triangle pointing toward the center. The missing 8th wedge must show that same small triangle, pointing toward the center, correctly oriented for its position in the circle — not pointing outward or at a random angle.

5. Border and Edge Pattern Continuation

When a figure has a decorative border or edge pattern running along its perimeter and the missing piece includes part of that border, the border pattern's spacing and shape must continue seamlessly across the gap at exactly the same rhythm as the border on either side of the missing piece.

- A rectangular figure has a dashed-line border with dashes evenly spaced every 1 cm running along the top edge. The missing piece covers a 3 cm stretch of that top edge. The correct piece must show exactly 3 dashes at 1 cm spacing, correctly positioned to align with the dashes immediately before and after the gap — a piece with only 2 dashes, or unevenly spaced dashes, breaks the border rhythm.

6. Checking the Piece's Outline Shape Matches the Gap

Before evaluating any internal pattern details, first confirm that a candidate piece's outer outline shape and size actually match the gap's outline exactly — a piece with perfect internal pattern-matching but a slightly wrong outline shape cannot be the correct answer regardless of how well its pattern seems to continue.

- The gap in a figure is a perfect right-angled rectangular notch. One candidate piece has beautifully matching internal stripes but has a very slightly rounded top-right corner. Despite the excellent internal pattern match, this piece is rejected on outline grounds alone — the gap requires a sharp right angle, not a rounded one.

7. Eliminating Options With Wrong Internal Details

Use the answer options as an elimination checklist just as in matrix questions: reject any option with an internal detail count, color, or shading that doesn't match what the surrounding pattern requires, even if only one small detail is off — exam distractors are commonly built to be correct in every way except one deliberately altered detail.

- The required piece must show 2 small dots and diagonal hatching. Option A has 2 dots but no hatching — eliminated. Option B has hatching but 3 dots instead of 2 — eliminated. Option C has exactly 2 dots and matching diagonal hatching — selected.

8. Worked Full Example: Completing a Symmetric Mandala Figure

Combine border, radial, and symmetry checks together for complex figures: first identify the overall symmetry or repetition type, then trace lines/borders into the gap, then filter candidate pieces by outline fit, and finally eliminate on internal detail mismatches.

- A circular mandala figure has 6-fold radial symmetry (6 identical wedges) AND each wedge is itself left-right symmetric. The missing wedge's design is derived from any of the other 5 intact wedges. A candidate piece matching that wedge's design but mirrored the wrong way (breaking the wedge's own internal left-right symmetry) is rejected, even though it matches the overall radial repetition — both layers of symmetry must be satisfied simultaneously.

Common Pitfalls

- Selecting a piece purely because its outline shape fits the gap, without checking whether its internal pattern actually continues correctly.
- Missing that a line entering the gap must exit at both the correct position AND the correct angle, not just anywhere on the far boundary.
- In radial (wheel/mandala) figures, using the wrong wedge count or angle, leading to an incorrectly oriented missing wedge.
- In border-pattern gaps, getting the dash/element count right but the spacing or alignment wrong relative to the border on either side.
- Accepting a piece with a slightly wrong outline shape because its internal pattern otherwise looks right.
- Checking only one layer of symmetry (radial OR mirror) in figures that actually require both to hold simultaneously.
- Not using full elimination against ALL confirmed details, stopping as soon as a piece matches just one or two of the required features.

Key Takeaways

- The missing piece must continue the figure's actual pattern (lines, repetition, symmetry), not just physically fit the gap's outline.

- For symmetric figures, the correct piece is the mirror image of the corresponding intact region across the symmetry line.
- Lines and curves must connect across the gap at both the correct position AND the correct angle.
- In radial/wheel figures, count total wedges and angle-per-wedge to determine the missing wedge's exact required orientation.
- Always verify outline shape and size fit BEFORE evaluating internal pattern details.
- Use full elimination against every confirmed detail (count, position, angle, shading), not just the first one or two that seem to match.

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IndiaBix — Non-Verbal Reasoning: Completion of Incomplete Pattern

<https://www.indiabix.com/non-verbal-reasoning/completion-of-incomplete-pattern/>

GeeksforGeeks — Non-Verbal Reasoning Questions and Answers

<https://www.geeksforgeeks.org/non-verbal-reasoning-questions-and-answers/>